

and

$$3p = -\frac{1}{3} \frac{b^2}{a^2} + \frac{c}{a} \quad \text{and} \quad 2q = \frac{2}{27} \frac{b^3}{a^3} - \frac{1}{3} \frac{bc}{a^2} + \frac{d}{a}.$$

Insertion leads to

$$3p = -\frac{71}{54} \frac{T^2}{L^2 m^2}, \quad 2q = -\frac{653}{1458} \frac{T^3}{L^3 m^3}.$$

From this, it follows that

$$q^2 + p^3 < 0,$$

i.e., there exist 3 real solutions of the cubic equation (8.39).

For the case $q^2 + p^3 \leq 0$, the solutions y_1, y_2, y_3 can be calculated using tabulated auxiliary quantities (see mathematical supplement 8.4). Direct application of Cardano's formula would lead to complex expressions for the real roots, hence the above method is convenient.

After insertion one obtains for the auxiliary quantities

$$\begin{aligned} \cos \varphi &= \frac{-q}{\sqrt{-p^3}}, & y_1 &= 2\sqrt{-p} \cos \frac{\varphi}{3}, \\ y_2 &= -2\sqrt{-p} \cos \left(\frac{\varphi}{3} + \frac{\pi}{3} \right), \\ y_3 &= -2\sqrt{-p} \cos \left(\frac{\varphi}{3} - \frac{\pi}{3} \right), \end{aligned}$$

and finally, for the eigenfrequencies of the system

$$\omega_1 = 0.563 \sqrt{\frac{T}{Lm}}, \quad \omega_2 = 0.916 \sqrt{\frac{T}{Lm}}, \quad \omega_3 = 1.585 \sqrt{\frac{T}{Lm}}.$$

Example 8.3: Exercise: Complicated coupled vibrational system

Determine the eigenfrequencies of the system of three equal masses suspended between springs with the spring constant k , as is shown in Figure 8.7. *Hint:* Consider the solution method of the preceding problem 8.2 and the mathematical supplement 8.4.

Solution From Figure 8.7, we extract for the equations of motion

$$\begin{aligned} m\ddot{x}_1 &= -kx_1 - k(x_1 - x_2) - k(x_1 - x_3), \\ m\ddot{x}_2 &= -kx_2 - k(x_2 - x_1) - k(x_2 - x_3), \\ m\ddot{x}_3 &= -kx_3 - k(x_3 - x_1) - k(x_3 - x_2), \end{aligned} \tag{8.40}$$

or

$$\begin{aligned} m\ddot{x}_1 + 3kx_1 - kx_2 - kx_3 &= 0, \\ m\ddot{x}_2 + 3kx_2 - kx_3 - kx_1 &= 0, \\ m\ddot{x}_3 + 3kx_3 - kx_1 - kx_2 &= 0. \end{aligned} \tag{8.41}$$

We look for the eigenvibrations. All mass points must vibrate with the same frequency. Thus, we adopt the *ansatz*

$$x_1 = A \cos(\omega t + \psi), \quad \ddot{x}_1 = -\omega^2 A \cos(\omega t + \psi),$$

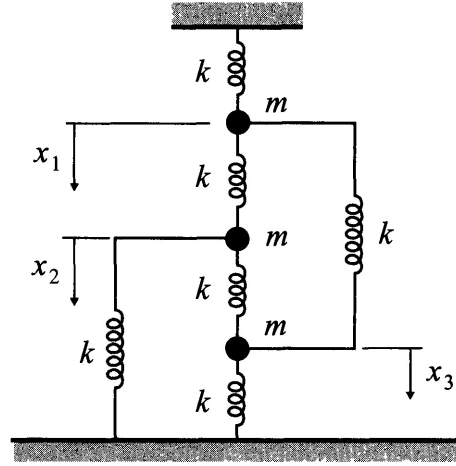


Figure 8.7. Vibrating coupled masses.

$$\begin{aligned} x_2 &= B \cos(\omega t + \psi), & \ddot{x}_2 &= -\omega^2 B \cos(\omega t + \psi), \\ x_3 &= C \cos(\omega t + \psi), & \ddot{x}_3 &= -\omega^2 C \cos(\omega t + \psi), \end{aligned}$$

and after insertion into equation (8.41), we get

$$\begin{aligned} (3k - m\omega^2)A - kB - kC &= 0, \\ -kA + (3k - m\omega^2)B - kC &= 0, \\ -kA - kB + (3k - m\omega^2)C &= 0. \end{aligned} \tag{8.42}$$

To get a nontrivial solution of equation (8.42), the determinant of coefficients must vanish:

$$\begin{vmatrix} (3k - m\omega^2) & -k & -k \\ -k & (3k - m\omega^2) & -k \\ -k & -k & (3k - m\omega^2) \end{vmatrix} = 0.$$

Expansion of the determinant leads to

$$0 = \omega^6 - \frac{9k}{m}\omega^4 + \frac{24k^2}{m^2}\omega^2 - \frac{16k^3}{m^3}$$

or

$$0 = \Omega^3 - \frac{9k}{m}\Omega^2 + \frac{24k^2}{m^2}\Omega - \frac{16k^3}{m^3},$$

where we substituted $\Omega = \omega^2$ (see Example 8.2). The general cubic equation $a\Omega^3 + b\Omega^2 + c\Omega + d = 0$ (in our case $a = 1$, $b = -9k/m$, $c = 24k^2/m^2$, $d = -16k^3/m^3$) can according to mathematical supplement 8.4 be reduced to

$$y^3 + 3py + 2q = 0,$$

where

$$y = \Omega + \frac{b}{3a}, \quad 3p = -\frac{1}{3} \frac{b^2}{a^2} + \frac{c}{a}, \quad 2q = \frac{2}{27} \frac{b^3}{a^3} - \frac{1}{3} \frac{bc}{a^2} + \frac{d}{a}.$$

Insertion leads to

$$3p = -3 \frac{k^2}{m^2}, \quad 2q = 2 \frac{k^3}{m^3} \quad \Rightarrow \quad q^2 + p^3 = 0,$$

i.e., there exist 3 solutions (the real roots); 2 of them coincide. Hence, the vibrating system being treated here is *degenerate*. As in problem 8.2, the solutions can be calculated using tabulated auxiliary quantities. For these, we obtain

$$\begin{aligned} \cos \varphi &= \frac{-q}{\sqrt{-p^3}}, & y_1 &= 2\sqrt{-p} \cos \frac{\varphi}{3}, \\ y_2 &= -2\sqrt{-p} \cos \left(\frac{\varphi}{3} + \frac{\pi}{3} \right), \\ y_3 &= -2\sqrt{-p} \cos \left(\frac{\varphi}{3} - \frac{\pi}{3} \right), \end{aligned}$$

and, after insertion, for the eigenfrequencies of the system

$$\omega_3 = \sqrt{\frac{k}{m}}, \quad \omega_1 = \omega_2 = 2\sqrt{\frac{k}{m}}.$$

Example 8.4: Mathematical supplement: The Cardano formula¹

In theoretical physics, one often meets the problem of solving a cubic equation, just as in the Examples 8.2 and 8.3. We now will clarify this problem.

Reduction of the general cubic equation: If the general cubic equation

$$x^3 + ax^2 + bx + c = 0 \tag{8.43}$$

with nonvanishing coefficients a , b , and c is to be solved, one must first eliminate the quadratic term of the equation, i.e., *reduce* the equation. If the unknown x is replaced by $y + \lambda$, where y and λ are new, unknown quantities, equation (8.43) turns into

$$\begin{aligned} (y^3 + 3y^2\lambda + 3y\lambda^2 + \lambda^3) + (ay^2 + 2ay\lambda + a\lambda^2) + (by + b\lambda) + c &= 0, \\ y^3 + (3\lambda + a)y^2 + (3\lambda^2 + 2a\lambda + b)y + (\lambda^3 + a\lambda^2 + b\lambda + c) &= 0. \end{aligned} \tag{8.44}$$

Since we have replaced one unknown quantity x by two unknown ones, y and λ , we can freely dispose of one of the two unknown quantities. This freedom is exploited so as to let the quadratic term of the equation disappear. This is achieved by setting the coefficient of y^2 , that is, $3\lambda + a$, equal to zero, i.e., $\lambda = -a/3$. By inserting this value the equation (8.44) changes to

$$y^3 + \left(-\frac{a^2}{3} + b \right) y + \left(\frac{2a^3}{27} - \frac{ab}{3} + c \right) = 0. \tag{8.45}$$

¹We follow the exposition of E. v. Hanxleben and R. Hentze, *Lehrbuch der Mathematik*, Friedrich Vieweg & Sohn 1952, Braunschweig-Berlin-Stuttgart.

If we set the expressions determined by the known coefficients a , b , and c of the cubic equation,

$$-\frac{a^2}{3} + b = p \quad \text{and} \quad \frac{2a^3}{27} - \frac{ab}{3} + c = q, \quad (8.46)$$

the cubic equation takes the form

$$y^3 + py + q = 0 \quad (\text{reduced cubic equation}). \quad (8.47)$$

Result: To reduce the cubic equation given in the normal form, one sets $x = y - a/3$. Then equation (8.47) follows from equation (8.43).

Example: $x^3 - 9x^2 + 33x - 65 = 0$.

(1) *Solution:* Set $x = y - (-3) = y + 3$.

$$\begin{aligned} (y+3)^3 - 9(y+3)^2 + 33(y+3) - 65 &= 0, \\ (y^3 + 9y^2 + 27y + 27) - 9(y^2 + 6y + 9) + 33(y+3) - 65 &= 0, \\ y^3 + 6y - 20 &= 0. \end{aligned}$$

(2) *Solution:* Insert the values calculated from equation (8.46) into equation (8.47).

Special case: If in the general cubic equation, the linear term is missing ($b = 0$), i.e., the cubic equation is given in the form

$$x^3 + ax^2 + c = 0, \quad (8.48)$$

the reduction can also be performed by inserting

$$x = \frac{c}{y}. \quad (8.49)$$

From equations (8.48) and (8.49), we obtain the reduced equation

$$\frac{c^3}{y^3} + a \frac{c^2}{y^2} + c = 0 \quad \text{or} \quad y^3 + acy + c^2 = 0. \quad (8.50)$$

Solution of the reduced cubic equation: If one sets in the reduced cubic equation

$$\begin{aligned} y^3 + py + q &= 0, \\ y &= u + v, \end{aligned} \quad (8.51)$$

one obtains

$$\begin{aligned} u^3 + 3u^2v + 3uv^2 + v^3 + p(u+v) + q &= 0, \\ (u^3 + v^3 + q) + 3uv(u+v) + p(u+v) &= 0, \\ (u^3 + v^3 + q) + (3uv + p)(u+v) &= 0. \end{aligned} \quad (8.52)$$

Since one can freely dispose of one of the unknown quantities u or v (justification?), these are suitably chosen so that the coefficient of $(u+v)$ vanishes. We therefore set

$$3uv + p = 0, \quad \text{i.e.,} \quad uv = -\frac{p}{3}. \quad (8.53)$$

Then equation (8.52) simplifies to

$$u^3 + v^3 + q = 0 \quad \text{or} \quad u^3 + v^3 = -q. \quad (8.54)$$

u and v are determined by equations (8.53) and (8.54). The quantities u and v can no longer be arbitrarily chosen. By raising equation (8.54) to the second power and equation (8.53) to the third power, one obtains

$$\begin{aligned} u^6 + 2u^3v^3 + v^6 &= q^2, \\ 4u^3v^3 &= -4\left(\frac{p}{3}\right)^3. \end{aligned}$$

Subtraction of the two equations yields

$$\begin{aligned} (u^3 - v^3)^2 &= q^2 + 4\left(\frac{p}{3}\right)^3, \\ u^3 - v^3 &= \pm \sqrt{q^2 + 4\left(\frac{p}{3}\right)^3}. \end{aligned} \quad (8.55)$$

By addition and subtraction of equations (8.54) and (8.55), one obtains

$$\begin{aligned} u^3 &= \frac{1}{2} \left[-q \pm \sqrt{q^2 + 4\left(\frac{p}{3}\right)^3} \right] \quad \text{and} \quad v^3 = \frac{1}{2} \left[-q \mp \sqrt{q^2 + 4\left(\frac{p}{3}\right)^3} \right], \\ u &= \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \quad \text{and} \quad v = \sqrt[3]{-\frac{q}{2} \mp \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}. \end{aligned} \quad (8.56)$$

If one sets

$$\sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} = m \quad \text{and} \quad \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} = n,$$

one gets

$$\begin{aligned} u_1 &= m, & u_2 &= m\epsilon_2, & u_3 &= m\epsilon_3, \\ v_1 &= n, & v_2 &= n\epsilon_2, & v_3 &= n\epsilon_3. \end{aligned}$$

Here, the ϵ_i are the unit roots of the cubic equation $x^3 = 1$ which, as is evident, read

$$\epsilon_1 = 1, \quad \epsilon_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad \epsilon_3 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

Since now $y = u + v$, one can actually form 9 values for y (why?). But since the quantities u and v must satisfy the determining equation (8.53), the number of possible connections between u and v is restricted to 3, namely,

$$y_1 = u_1 + v_1, \quad y_2 = u_2 + v_3, \quad y_3 = u_3 + v_2;$$

hence,

$$\begin{aligned} y_1 &= m + n = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}, \\ y_2 &= m\epsilon_2 + n\epsilon_3 = -\frac{m+n}{2} + \frac{m-n}{2}i\sqrt{3}, \\ y_3 &= m\epsilon_3 + n\epsilon_2 = -\frac{m+n}{2} - \frac{m-n}{2}i\sqrt{3}. \end{aligned} \quad (8.57)$$

The real root of the cubic equation, i.e., the root

$$y_1 = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}$$

is known as the “*Cardano formula*.” It was named in honor of the Italian Hieronimo Cardano² to whom the discovery of the formula was falsely ascribed. Actually, the formula is due to the Bolognesian professor of mathematics Scipione del Ferro,³ who found this ingenious algorithm.

Example: $y^3 - 15y - 126 = 0$. Here,

$$\begin{aligned} p &= -15, & q &= -126, \\ \frac{p}{3} &= -5, & \frac{q}{2} &= -63. \end{aligned}$$

By inserting into the Cardano formula, one obtains

$$\begin{aligned} y_1 &= \sqrt[3]{63 + \sqrt{63^2 - 5^3}} + \sqrt[3]{63 - \sqrt{63^2 - 5^3}} \\ &= \sqrt[3]{63 + \sqrt{3844}} + \sqrt[3]{63 - \sqrt{3844}} \\ &= \sqrt[3]{63 + 62} + \sqrt[3]{63 - 62} \\ &= \sqrt[3]{125} + \sqrt[3]{1} \quad (= m + n) \\ &= 6, \\ y_2 &= -\frac{5+1}{2} + \frac{5-1}{2}i\sqrt{3} = -3 + 2i\sqrt{3}, \\ y_3 &= -\frac{5+1}{2} - \frac{5-1}{2}i\sqrt{3} = -3 - 2i\sqrt{3}. \end{aligned}$$

Check the validity of the roots by insertion!

Discussion of Cardano’s formula: The square root appearing in the Cardano formula only yields a real value if the radicand $(q/2)^2 + (p/3)^3 \geq 0$. If the radicand is negative, the three values for y yield complex numbers. We consider the possible cases:

²Hieronimo Cardano, Italian physicist, mathematician, and astrologer, b. Sept. 24, 1501, Pavia–d. Sept. 20, 1576, Rome. Cardano was the illegitimate son of Fazio (Bonifacius) Cardano, a friend of Leonardo da Vinci. He studied at the universities of Pavia and Padua, and in 1526 he graduated in medicine. In 1532, he went to Milan, where he lived in deep poverty, until he got a position teaching in mathematics. In 1539, he worked at a high school of physics, where he soon became the director. In 1543, he accepted a professorship for medicine in Pavia.

As a mathematician, Cardano was the most prominent personality of his age. In 1539, he published two books on arithmetic methods. At this time, the discovery of a solution method for the cubic equation became known. Nicolo Tartaglia, a Venetian mathematician, was the owner. Cardano tried in vain to get permission to publish it. Tartaglia left the method to him under the condition that he keeps it secret. In 1545, Cardano’s book *Artis magnae sive de regulis algebraicis*, one of the cornerstones of the history of algebra, was published. The book contained, besides many other new facts, the method of solving cubic equations. The publication caused a serious controversy with Tartaglia.

³Scipione del Ferro, b. 1465(?)–d. 1526 (?). About his life we know only that he lectured from 1496 to 1526 at the university of Bologna. By 1500, he discovered the method of solving the cubic equation but did not publish it. Tartaglia rediscovered the method in 1535.

		$\sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}$	Form of the roots
(1)	$p > 0$	Real	A real value, two complex conjugate values
(2)	$p < 0$, namely,		
(a)	$\left \left(\frac{p}{3}\right)^3\right < \left(\frac{q}{2}\right)^2$	Real	As in 1.
(b)	$\left \left(\frac{p}{3}\right)^3\right = \left(\frac{q}{2}\right)^2$	$= 0$	Three real values, among them a double root
(c)	$\left \left(\frac{p}{3}\right)^3\right > \left(\frac{q}{2}\right)^2$	Imaginary	All three roots by the form imaginary

The case (2c) was of particular interest to the mathematicians of the Middle Ages. Since any cubic equation has at least one real root, but they could not find it by means of Cardano's formula, the case was called the *casus irreducibilis*.⁴ The first to solve this case was the French politician and mathematician Vieta.⁵ He proved by using trigonometry that this case was solvable too, and that in this case the equation has three real roots.

Trigonometric solution of the irreducible case: Since p is negative in this case, one starts from the reduced cubic equation

$$y^3 - py + q = 0, \quad (8.58)$$

where p must now be kept fixed as absolute numerical value. According to the trigonometric formulae we have

$$\begin{aligned}
 \cos 3\alpha &= \cos(2\alpha + \alpha) = \cos 2\alpha \cos \alpha - \sin 2\alpha \sin \alpha \\
 &= (\cos^2 \alpha - \sin^2 \alpha) \cos \alpha - 2 \sin^2 \alpha \cos \alpha \\
 &= \cos^3 \alpha - \sin^2 \alpha \cos \alpha - 2 \sin^2 \alpha \cos \alpha \\
 &= \cos^3 \alpha - (1 - \cos^2 \alpha) \cos \alpha - 2(1 - \cos^2 \alpha) \cos \alpha \\
 &= \cos^3 \alpha - \cos \alpha + \cos^3 \alpha - 2 \cos \alpha + 2 \cos^3 \alpha \\
 &= 4 \cos^3 \alpha - 3 \cos \alpha,
 \end{aligned}$$

⁴*Casus irreducibilis* (Lat.) = "the nonreducible case."

⁵*Francois Vieta*, French mathematician, b. 1540, Fontenay-le-Comte-d. Dec. 13, 1603, Paris. Advocate and advisor of Parliament in the Bretagne. His greatest achievements were in the theory of equations and algebra, where he introduced and systematically used letter notations. He established the rules for the rectangular spherical triangle which are often ascribed to Neper. In his *Canon mathematicus*, a table of angular functions (1571), he emphasized the advantages of decimal notation. [BR]

thus,

$$\cos^3 \alpha - \frac{3}{4} \cos \alpha - \frac{1}{4} \cos 3\alpha = 0. \quad (8.59)$$

If one considers $\cos \alpha$ to be unknown, equation (8.59) coincides with the form of equation (8.58). But since the value of the cosine varies only between the limits -1 and $+1$, while y , according to the values of p and q , can take any values, one cannot simply set $\cos \alpha = y$. By multiplying equation (8.59) by a still uncertain positive factor ϱ^3 , one obtains

$$\varrho^3 \cos^3 \alpha - \frac{3}{4} \varrho^2 \cdot \varrho \cos \alpha - \frac{1}{4} \varrho^3 \cos 3\alpha = 0. \quad (8.60)$$

By setting $\varrho \cdot \cos \alpha = y$, $p = (3/4)\varrho^2$, and $q = -(1/4)\varrho^3 \cos 3\alpha$, equation (8.60) turns into (8.58). From this, we find

$$\varrho = 2 \cdot \sqrt{\frac{p}{3}} \quad (8.61)$$

and

$$\cos 3\alpha = -\frac{4q}{\varrho^3} = \frac{-4q}{8 \cdot (p/3)\sqrt{p/3}} = -\frac{q/2}{\sqrt{(p/3)^3}}. \quad (8.62)$$

Equation (8.62) is ambiguous, since the cosine is a periodic function. One has

$$3\alpha = \varphi + k \cdot 360^\circ, \quad \text{where } k = 0, 1, 2, 3, \dots \quad (8.63)$$

From this, we find for α

$$\alpha_1 = \frac{\varphi}{3}, \quad \alpha_2 = \frac{\varphi}{3} + 120^\circ, \quad \alpha_3 = \frac{\varphi}{3} + 240^\circ.$$

Compare this consideration with the problem of cyclotomy! Which values are obtained for α if $k = 3, 4, \dots$?

For y , one obtains

$$y_1 = 2\sqrt{\frac{p}{3}} \cos \frac{\varphi}{3}, \quad y_2 = 2\sqrt{\frac{p}{3}} \cos \left(\frac{\varphi}{3} + 120^\circ \right), \quad y_3 = 2\sqrt{\frac{p}{3}} \cos \left(\frac{\varphi}{3} + 240^\circ \right).$$

Now

$$\cos \left(\frac{\varphi}{3} + 120^\circ \right) = -\cos \left(60^\circ - \frac{\varphi}{3} \right)$$

and

$$\cos \left(\frac{\varphi}{3} + 240^\circ \right) = -\cos \left(60^\circ + \frac{\varphi}{3} \right),$$

so that the roots of the cubic equations are

$$\begin{aligned} y_1 &= 2\sqrt{\frac{p}{3}} \cos \frac{\varphi}{3}, \\ y_2 &= -2\sqrt{\frac{p}{3}} \cos \left(60^\circ - \frac{\varphi}{3} \right), \\ y_3 &= -2\sqrt{\frac{p}{3}} \cos \left(60^\circ + \frac{\varphi}{3} \right). \end{aligned} \quad (8.64)$$

Comment: The formulas of the *casus irreducibilis* can also be derived by means of the Moivre's theorem.

Example: Calculate the roots of the equation

$$y^3 - 981y - 11340 = 0.$$

Solution: Since $p < 0$ and

$$\left| \left(\frac{p}{3} \right)^3 \right| = 327^3, \quad \log \left| \left(\frac{p}{3} \right)^3 \right| = 3 \cdot \log 327 = 7.5436,$$

$$\left(\frac{q}{2} \right)^2 = 5670^2, \quad \log \left(\frac{q}{2} \right)^2 = 2 \cdot \log 5670 = 7.5072,$$

by comparing the logarithms it follows that $|(p/3)^3| > (q/2)^2$. Thus, the condition of the *casus irreducibilis* is fulfilled. According to equation (8.62)

$$\cos 3\alpha = + \frac{5670}{\sqrt{327^3}},$$

$$\log \cos 3\alpha = 3.7536 - 3.7718 = 9.9818 - 10,$$

$$\varphi = 3\alpha \approx 16^\circ 30', \quad \text{hence,} \quad \frac{\varphi}{3} = \alpha = 5^\circ 30'.$$

From equation (8.64), we obtain $y_1 = 36$, $y_2 = -21$, $y_3 = -15$. Check the root values by insertion!
